



History of Git: A Beginner's Guide

Introduction

What is Git?

- **Git** is a version control system that helps developers track changes in their code over time.
- It allows multiple people to work on a project simultaneously without conflicts.
- **Used by:** Software developers, open-source projects, and companies for collaboration.

Why Was Git Created?

- Before Git, developers used **BitKeeper**, a proprietary version control system.
- In **2005**, BitKeeper's free license for Linux developers was revoked.
- **Linus Torvalds**, the creator of Linux, decided to build a **new, free, and powerful version control system**.
- Thus, **Git was born!**

Timeline: Evolution of Git

2005: The Birth of Git

- Linus Torvalds designed **Git** for managing the Linux kernel's source code.
- Goals:
 - **Speed**
 - **Distributed development** (no central server required)
 - **Strong support for non-linear development** (branches & merging)



- **Security & integrity**

2006-2008: Early Adoption

- **2006:** Git becomes **stable** and widely used in Linux development.
- **2007:** GitHub is launched, making Git adoption easier.
- **2008:** Major companies start using Git for software development.

2010s: Git Dominates the Industry

- **2013:** Git becomes the most popular version control system.
- **2018:** Microsoft acquires GitHub for **\$7.5 billion**, proving Git's dominance.
- **2020s:** Git remains the **standard** for software development, used by **millions of developers worldwide**.

How Git Works: Basic Concepts

1. Repository (Repo)

- A **repository** is like a project folder where all code and history are stored.
- It tracks **every change** made to the files.

2. Commit

- A **commit** is a snapshot of the changes made to the code.
- Every commit has a unique **ID (hash)**.

3. Branching & Merging

- **Branching** allows developers to work on different features without affecting the main project.
- **Merging** combines different branches into one.



Installing Git

Windows

1. Download **Git for Windows** from git-scm.com.
2. Install and open **Git Bash**.
3. Verify installation:

```
git --version
```

Linux (Ubuntu)

```
sudo apt update  
sudo apt install git -y  
git --version
```

MacOS

```
brew install git  
git --version
```

Basic Git Commands

Check Git Version

```
git --version
```

Set Up User Info

```
git config --global user.name "Your Name"  
git config --global user.email "your_email@example.com"
```



Initialize a Git Repository

```
git init
```

Clone a Repository

```
git clone <repository_url>
```

Add Changes to Staging Area

```
git add .
```

Commit Changes

```
git commit -m "Your commit message"
```

Push Changes to Remote Repository

```
git push origin main
```

Pull Latest Changes

```
git pull origin main
```

Know More: FAQs

1. Why is Git called a "distributed" version control system?

- Unlike older systems, Git does not require a central server. Every developer has a full copy of the repository.

2. How is Git different from GitHub?

- **Git** is the version control tool.



- **GitHub** is a platform to host Git repositories online.

3. What is the purpose of a commit?

- A **commit** saves the current changes in the project with a unique ID, so you can track progress or roll back if needed.

4. How do I check the history of commits?

```
git log
```

5. How can I undo a commit?

```
git revert <commit_id>
```

Conclusion

Git revolutionized **software development** by providing a fast, reliable, and distributed version control system. Understanding Git will help you collaborate efficiently and manage projects like a pro!